Subject Description Form

Subject Code	ME5208
Subject Title	Energy and Climate Change
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To understand the scientific principles underlying various energy systems and climate change. 2. To analyze the carbon footprint and climate impact of different energy technologies. 3. To evaluate emerging energy conservation technologies and their effectiveness in mitigating climate change. 4. To develop critical thinking and technical skills on the sustainability of energy systems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand the principles of various energy systems and their contribution to climate change. b. Understand the physics of climate change, its causes and potential consequences. c. Apply fundamental knowledge to assess the climate impacts of various energy systems. d. Propose sustainable solutions for energy challenges in the context of climate change.
Subject Synopsis/ Indicative Syllabus	***Introduction to climate change:*** overview of climate science; natural carbon cycle; greenhouse effects; causes and potential consequences of climate change. ***Energy conversion and storage technologies:*** renewable and non-renewable energy sources; traditional energy conversion systems; renewable energy conversion and storage systems. ***Energy and carbon audit:*** local and international guidelines in energy and carbon audit; carbon accounting of various energy conversion, storage and conservation technologies; carbon footprint calculator; carbon tax. ***Decarbonization technologies:*** efficiency enhancement methods; energy conservation practices in the commercial and industrial sectors; carbon capture and storage technologies. ***Case study***: analyse life cycle carbon audit of energy systems; propose and evaluate the effectiveness of carbon taxes.

Teaching/Learning Methodology	The realization of the intended learning outcomes will be primarily based on lectures under adequate guidance from subject instructors. Students will also be directed to complete a team project with report and presentation to enhance understanding of the subject contents and practice presentation skills.
Assessment Methods in Alignment with Intended Learning Outcomes	(see table below)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d		
1. Assignment	30	✓	✓	✓	✓		
2. Project	30	✓	✓	✓	✓		
3. Examination	40	✓	✓				
Total	100 %						

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Overall Assessment:

0.40 × Examination + 0.60 × Continuous Assessment

1. The continuous assessment will comprise two components: team project (30%) and individual assignment (30%). The team project and assignment are aimed at evaluating their understanding on the physical principles, formulating and solving related problems and enhancing the computational skills.

2. The examination (40%) will be used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently, and to determine the degree of achieving the subject learning outcomes.

Student Study Effort Expected		
	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Project presentation	6 Hrs.
	Other student study effort:	
	▪ Self-learning	36 Hrs.
	▪ Assignments and projects	50 Hrs.
	Total student study effort	125 Hrs.

Reading List and References	Mathez, Edmond, and Jason Smerdon. *Climate change: the science of global warming and our energy future*. Columbia University Press, 2018. MacKay, David JC. Sustainable Energy-without the hot air. Bloomsbury Publishing, 2016. Everett, Robert, Godfrey Boyle, Stephen Peake, and Janet Ramage. *Energy systems and sustainability: power for a sustainable future*. Oxford University Press, 2012. Harvey, Hal, Robbie Orvis, Jeffrey Rissman, Michael O'Boyle, Chris Busch, and Sonia Aggarwal. *Designing climate solutions: a policy guide for low-carbon energy*. Washington: Island Press, 2018.
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